How much solute is there? Concentration

Concentration in g/L

$$C = \frac{\text{mass of solute (g)}}{\text{volume of solution (L)}}$$

grams of solute in each litre of solution



15 g of salt → 0.5 L of solution

$$C = \frac{15\ g}{0.5\ L} = \mathbf{30\ g/L}$$

Percentage by mass

$$\%\ \text{mass} = \frac{\text{mass of solute (g)}}{\text{mass of solution (g)}} \cdot 100$$

grams of solute in each 100 g of solution



10 g of sugar + 90 g of water = 100 g

$$\% = \frac{10\ g}{100\ g} \cdot 100 = \mathbf{10\ \%}$$

Percentage by volume

$$\%\ \text{vol} = \frac{\text{volume of solute}}{\text{volume of solution}} \cdot 100$$

mL of solute in each 100 mL of solution



12 mL of alcohol + 88 mL of water = 100 mL

$$\% = \frac{12\ mL}{100\ mL} \cdot 100 = \mathbf{12\ \%}$$

⚠ Below the line there is always the **SOLUTION**, not the solvent: **solution = solute(s) + solvent**